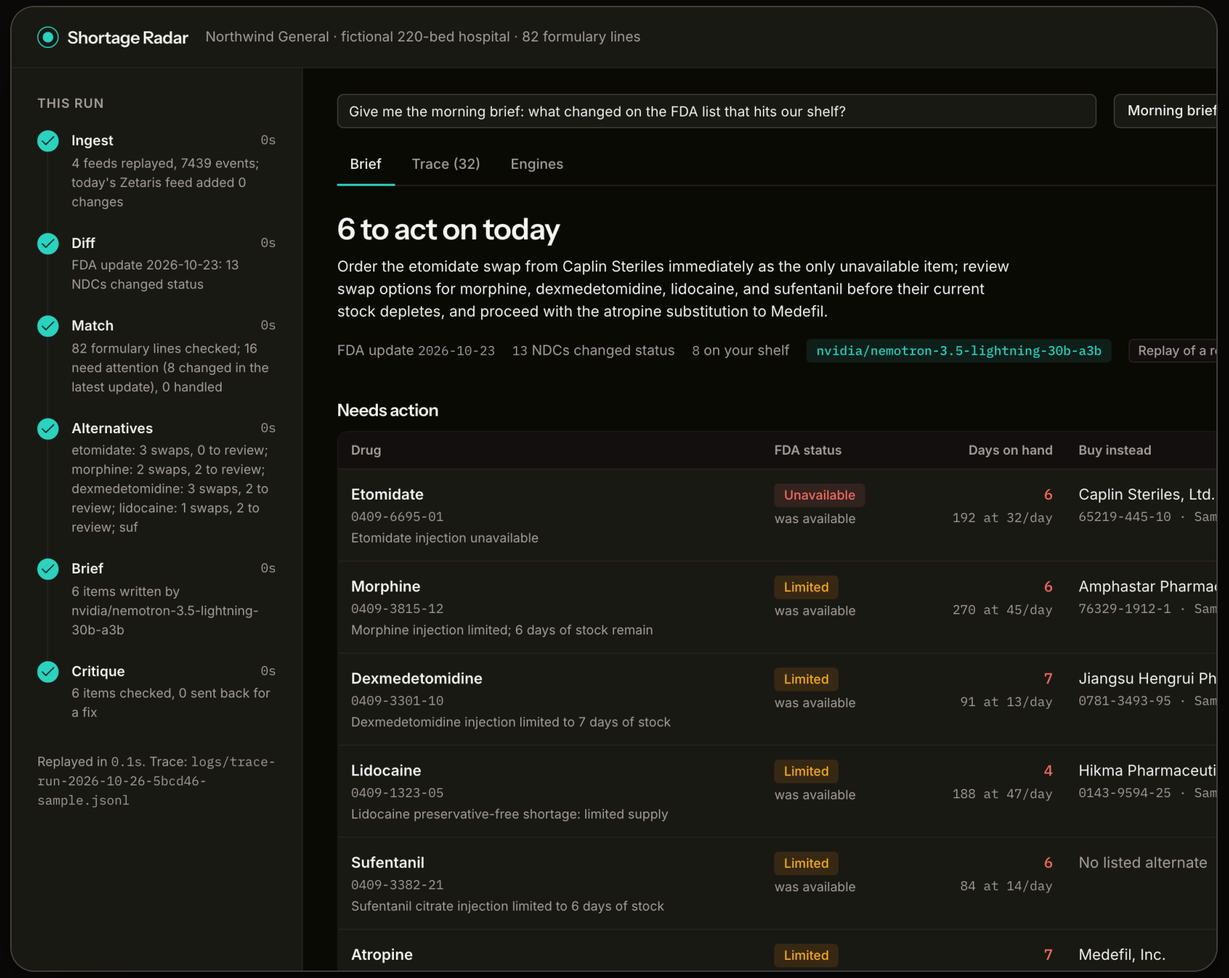
OPEN AGENT HACKATHON 2026 · TRACK 02 · THE AGENT THAT CAN EXPLAIN WHY

Shortage Radar

It doesn’t guess what will run short. It tells a hospital pharmacy buyer what to do about what just did.

Zetaris · World Model · H-MEM · Scout Intent · Meterless Nightshift · NVIDIA Nemotron



Solo build · github.com/techbone/shortage-radar

Every morning the FDA shortage list moves. Which changes are yours?

1 **The user**

The buyer at a community hospital pharmacy. No supply-chain analytics team.

2 **What they do today**

Check the FDA and ASHP pages, read wholesaler allocation emails, keep a spreadsheet. Often they learn a drug is short when an order fails.

3 **What it costs**

Hours of labor, and the risk of buying the wrong substitute: a different concentration is a medication error.

746 /1,609

FDA shortage entries updated in one week

measured by us, week to 22 Oct 2026

8.6M

extra labor hours a year managing shortages, about $360M

Vizient survey of US hospitals, 2019

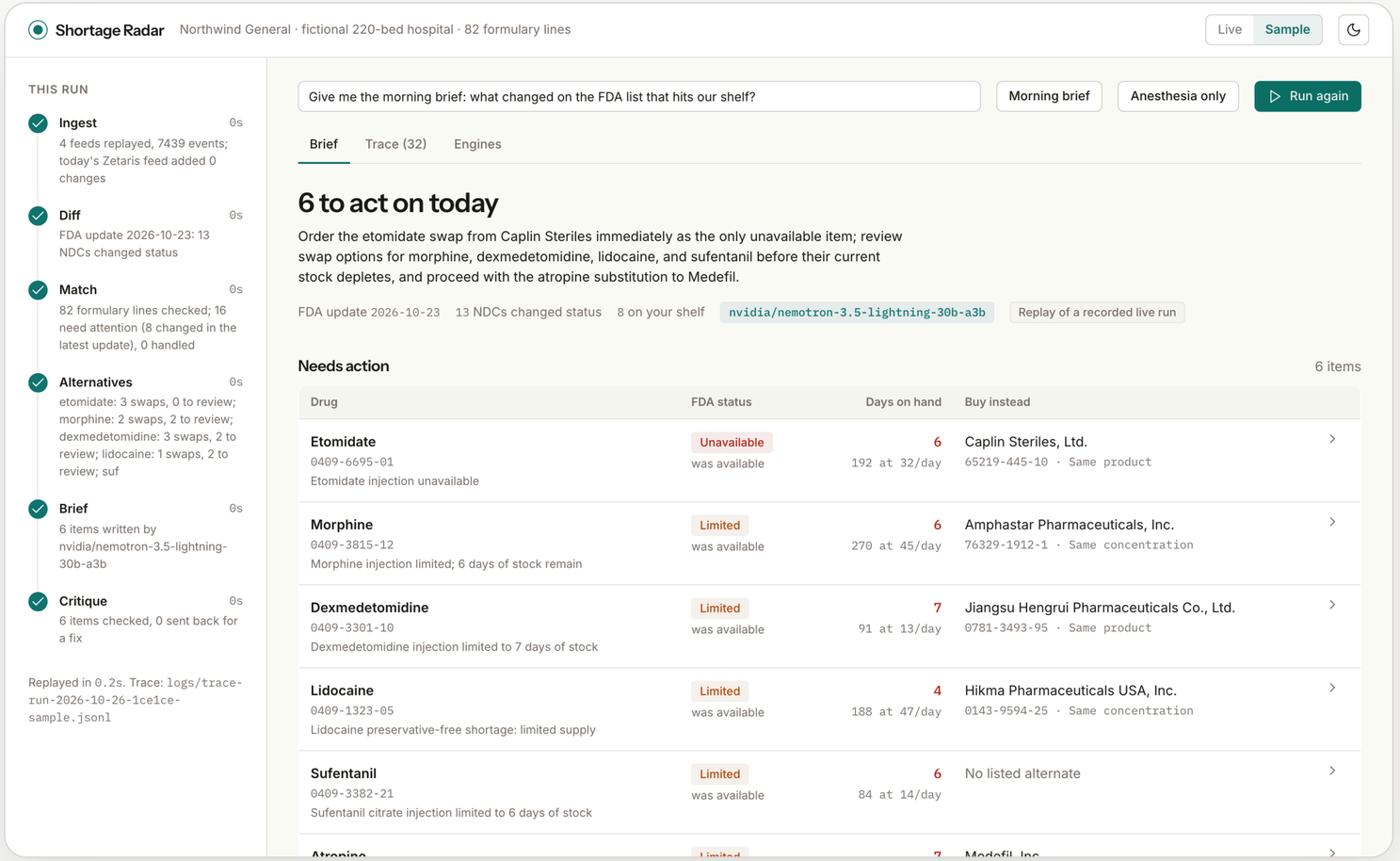
323

active shortages in Q1 2024, a record since 2001

ASHP

THE SOLUTION

One brief: what hit your shelf, how long it lasts, what to buy instead



1 **Matches stock to the FDA list**
One SQL statement joins the hospital's database with the live feed.

2 **Explains every line**
Each sentence cites the query, record or memory behind it.

3 **Remembers decisions**
Handled items stay quiet until the FDA status moves.

Real FDA data and NDCs. Fictional hospital: "Northwind General".

AGENT WORKFLOW

Six steps, five roles, one rule: code checks the model

01

Ingest

Zetaris + World Model

Read the FDA feed. Replay every saved snapshot.

02

Diff

World Model

What changed since the last update, with dated history.

03

Match

Zetaris · Scout · H-MEM

Join stock and FDA status. Screen the rows. Set aside what is handled.

04

Alternatives

Planner · Nemotron

Choose what to cover. Find same-strength products from other makers.

05

Brief

Writer · Nemotron

Write one item at a time, citing evidence ids.

06

Critique

Critic · plain code

Reject any claim its evidence does not back.

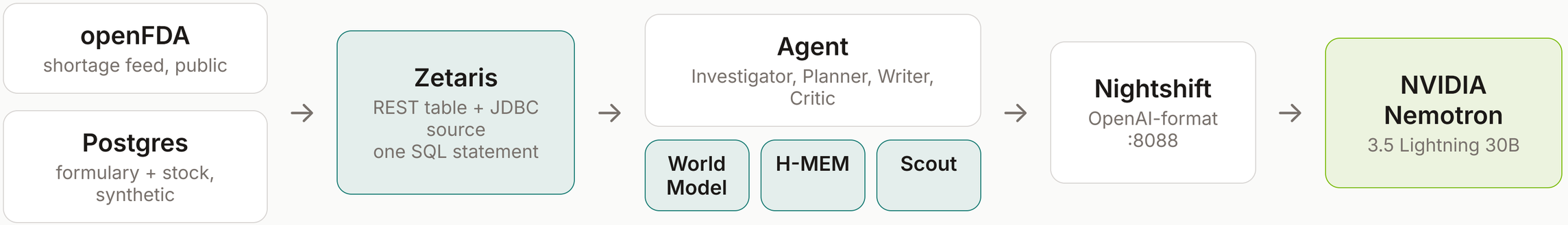
When the Critic rejects a draft

The Writer gets a fresh prompt with the reasons, up to twice. Still wrong? That item becomes plain, deterministic text, labelled as such.

Bounded by design

5-minute run budget, timeouts on every call, two model calls at a time. If Zetaris or Nightshift is unreachable, the run falls back to a recorded replay and says so.

Zetaris is the data layer. Nightshift is the only AI provider.



App

TypeScript, Node 22, Next.js 15, React 19. One port serves the UI and `POST /run`.

Data

Zetaris Lightning SQL, Postgres 16, saved FDA snapshots, recorded fixtures for offline replay.

Shipping

Docker image built once, secrets only at run time, 76 tests, JSONL trace per run.

Two sources, one statement, every number traceable

```
-- hospital Postgres (JDBC) joined with the FDA feed (REST)
SELECT f.ndc, f.generic_name,
       i.on_hand_units, i.daily_usage_units,
       s.availability AS fda_availability
FROM northwind_pg.formulary_item f
JOIN northwind_pg.inventory i ON i.ndc = f.ndc
LEFT JOIN fda.shortages s
  ON s.package_ndc = f.ndc
 AND s.status <> 'To Be Discontinued'
```

1. **FDA feed as a REST table**
   `CREATE LIGHTNING REST TABLE`, cached, flattened. 1,611 rows, equal to openFDA's own total.
2. **Hospital DB as a JDBC source**
   `CREATE DATASOURCE … jdbcurl`. The syntax came from the parser's error messages.
3. **Read-only by construction**
   Stacked statements and any write are refused before they reach Zetaris.

3 queries per run, about 1.5 s each

HOW THE METERLESS ENGINES ARE USED

Three engines, each in a feature you can click, each with a test that passes

World Model

8/8

official conformance checks

Replays every FDA snapshot. Powers “what changed” and the dated status history.

H-MEM

13/13

official conformance checks

Stores what the buyer decided. Handled items leave the brief and return if the FDA status moves.

Scout Intent

1.000 / 0.980

injection precision / recall, 2,316 cases

Screens FDA notes and database rows before any model reads them. Withholds only the poisoned field.

Official runners from Meterless, unmodified, checksummed in the repo. Honest caveat: on 6 attacks written after our rules, Scout caught 0. We patched 5. Polish still gets through.

docs/evidence/ · npm run conformance:wm · conformance:hmem · evals:injection

Nemotron plans and writes. The model is named on every call.

```
trace-example-1.jsonl (a real run)
+ 1.9s  Zetaris   fda-feed     1,611 rows
+ 2.6s  Guard     scout:screen 82 rows screened
+ 2.6s  Planner   plan:start   thinking on
+41.9s  Planner   plan         nemotron-3.5-lightning-30b-a3b
+43.1s  Zetaris   alternates   464 rows
+45.9s  Writer    write        nemotron-3.5-lightning-30b-a3b
+45.9s  Critic    critique     every claim checks out
+48.3s  Writer    write        nemotron-3.5-lightning-30b-a3b
+48.3s  Critic    critique     every claim checks out
   …
+58.9s  done      6 items, success
```

Thinking is a design lever

One-line answer, thinking on

```
≈ 4.3 s · 389 tokens
```

Same answer, thinking off

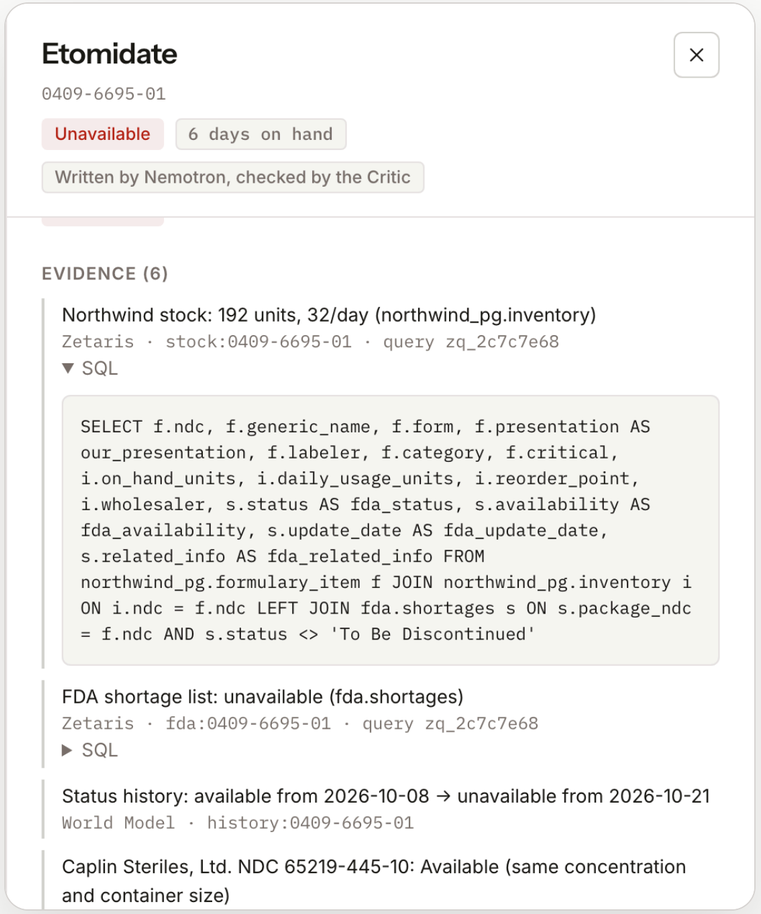
```
0.8 s · 23 tokens
```

The **Planner** keeps thinking on: choosing what matters is the judgement call, and it takes 39 s of a 59 s run. The **Writer** turns it off, 2 to 6 s per item, because the Critic checks every claim anyway.

First live request failed with an empty reply: an 80-token cap had been spent on thinking.

WHY IT CAN BE TRUSTED

Every sentence has a receipt, and a checker that is not a model



The Critic rejects a draft when:

the recommended NDC is not a same-concentration swap

it does not cite its own alternate's record

a cited evidence id does not exist

days on hand do not match the stock data

it names a cause the FDA note does not state

it predicts a shortage

Caught in a live run: the Writer cited another drug's alternate (`alt:76329-1912-1`) on sufentanil. The Critic sent it back; the retry was clean.

tests/critic.test.ts · src/agent/brief.ts

WHAT WAS UNIQUE

Sponsor technology doing the work, and a result we were willing to throw away

Remove any sponsor, the flow stops

No Zetaris: no join of private stock and public feed. No World Model: no “what changed”. No H-MEM: handled items re-alert daily. No Scout: planted text reaches the model. No Nemotron: no plan, no write-ups.

We tested our first idea and dropped it

Do recalls predict shortages? Over 142 shortages, they were *less* common before one (0.7–0.9× baseline). So the agent never forecasts. It reports exposure, runway and swaps.

A swap means the same strength

The obvious query offered 10 mg/mL for 20 mg/mL and a premixed drip for a vial. Strengths are parsed to per-mL; combinations and preservative mismatches go to pharmacist review.

The checker is code

A second model agreeing with the first proves little. A function that recomputes the NDC, the citation and the day count cannot be talked into agreeing.

What it does not do yet

Limits, stated plainly

The hospital is fictional: real NDCs, invented stock. No patient data.

Scout is pattern-based and covers the guard stage only.

Strengths are read from FDA presentation text. Anything it cannot parse goes to review, never to a swap.

The free NVIDIA endpoint has latency spikes. The adapter retries once; the Critic loop falls back to plain text.

Decision support only. The buyer confirms with the label and the wholesaler.

Next

Wholesaler availability and NADAC prices as more Zetaris sources.

A Nemotron safety model as Scout’s second, model-based layer.

ASHP and manufacturer notices beside the FDA list.

Daily brief by email, and multi-hospital tenancy.

Real-hospital pilot on read-only inventory exports.

TRY IT IN TWO MINUTES, NO KEYS

Shortage Radar doesn’t guess. It tells you what to do about what just went short.

```
# Sample mode needs no Zetaris, Nightshift or network
docker build -t oah-submission . && docker run --rm -p 8000:8000 oah-submission
# then open http://localhost:8000 and press Run sweep
```

github.com/techbone/shortage-radar

76 tests

World Model 8/8

H-MEM 13/13

Scout 1.000 / 0.980

nemotron-3.5-lightning-30b-a3b